

x''

$$ax^2 + bx' + c \cdot x^0 = 0$$

$$ax^2 + bx + c = 0$$

↓

$$x^3 + 2x^2 + 3x + 1 = 0$$

↓

$$x^3 + 2 \left(\frac{1}{x^2} \right) + 3x + 1 = 0$$

x

$$x^3 + 2 \cdot x^{-2}$$

x^2

(4a) $\rightarrow \boxed{ax^2 + bx + c = 0} \rightarrow$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$\Rightarrow \cancel{4a^2x^2} + \boxed{4abx} + \cancel{4ac} = 0 \leftarrow$

$\Rightarrow \frac{(2ax)^2}{a} + \frac{2 \cdot 2 \cdot ax \cdot b}{a} + \frac{b^2}{a} - \frac{b^2}{a} + \frac{4ac}{a} = 0$

$\Rightarrow (2ax + b) \boxed{-b + 4ac} = 0$

$\boxed{-\frac{b \pm \sqrt{b^2 - 4ac}}{2a}}$

$\Rightarrow (2ax + b)^2 = \boxed{b^2 - 4ac}$

$2ax + b = \pm \sqrt{b^2 - 4ac}$

$2ax = \boxed{-b} \pm \sqrt{b^2 - 4ac}$

$\Rightarrow \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$
 $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

$$\begin{array}{|c|} \hline 5 \\ \hline \end{array} \quad \begin{array}{|c|} \hline \sqrt{2} \\ \hline \end{array} \rightarrow \text{ଅବଶ୍ୟକ} \quad \begin{array}{|c|} \hline \sqrt{2} \\ \hline \end{array} = \begin{array}{|c|} \hline 1.3 \dots \\ \hline \end{array}$$

$$\frac{p}{q}$$

$$\begin{array}{|c|} \hline 5 + \sqrt{4} \\ \hline 2 \\ \hline \end{array}$$

$$\sqrt{4} = 2$$

ଅବଶ୍ୟକ

$$\begin{array}{|c|} \hline 6 - 4ac \\ \hline \end{array}$$

ଅବଶ୍ୟକ ଅବଶ୍ୟକ
ଅବଶ୍ୟକ

$$\frac{5 + 2}{2} = \left(\frac{7}{2} \right) \frac{p}{q}$$

x →

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x \rightarrow$

$$\frac{5 + \sqrt{2}}{2} \rightarrow D$$

$$\frac{p}{2}$$

$$\sqrt{-(ve)}$$

$$\frac{-b}{2a}$$

$$\frac{-b}{2a}$$

$$b^2 - 4ac < 0$$

$$b^2 - 4ac \geq 0$$

$$\frac{p}{q}$$

$$\frac{3}{2} \rightarrow p$$

1.32

~~(i) 27 34 4~~
(ii) 27 34 4
(iii) 27 34

$$\sqrt{3} = 1.3 \dots \dots$$

27 57 ✓
 $X = 4575$

(1) 27 57 ✓
(2) 27 57 ✓
(3) 27 57 ✓

$$\textcircled{4} \pm$$

$$4 + 0 = \textcircled{4}$$

$$4 - 0 = \textcircled{4}$$

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$[b^2 - 4ac] = 0$$

$$ax^2 + bx + c = 0$$

225

$$x = \alpha$$

or

$$x = \beta$$

$$\Rightarrow (x - \alpha)(x - \beta) = 0$$

$$\Rightarrow x^2 - \alpha x - \beta x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

~~***~~ x^2

($\alpha + \beta$) x + $\alpha\beta$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

α, β

x - ମୂଳ ସମୀକରଣ

α, β

(α, β)

$$ax^2 + bx + c = 0$$

$$\frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} = 0$$

$$\Rightarrow x^2 - \left(-\frac{b}{a}\right)x + \frac{c}{a} = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

** ১। p এর মান কত হলে $px^2 + 3x + 4 = 0$ সমীকরণের বীজগুলো (i) জটিল (ii) সমান (iii) বাস্তব ও অবাস্তব হবে?

বাকশির্ষো- ১৯৮৭, ৮৮, ৮৯, ২০০১

$$ax^2 + bx + c$$

$$D < 0$$

$$D > 0$$

$$D = b^2 - 4ac$$

$$= 3^2 - 4p \cdot 4$$

$$= 9 - 16p$$

$$= 9 - 16p$$

$$= 9 - 9$$

$$= 0 \quad (-ve)$$

$$9 - 16p = 0$$

$$9 - 16p = -9$$

$$-16 = -9$$

$$-16p = -9$$

$$\sqrt{b^2 - 4ac}$$

$$D = \sqrt{b^2 - 4ac}$$

$$\sqrt{-ve} = \text{জটিল}$$

$$D < 0$$

$$\sqrt{-ve} \rightarrow \text{জটিল}$$

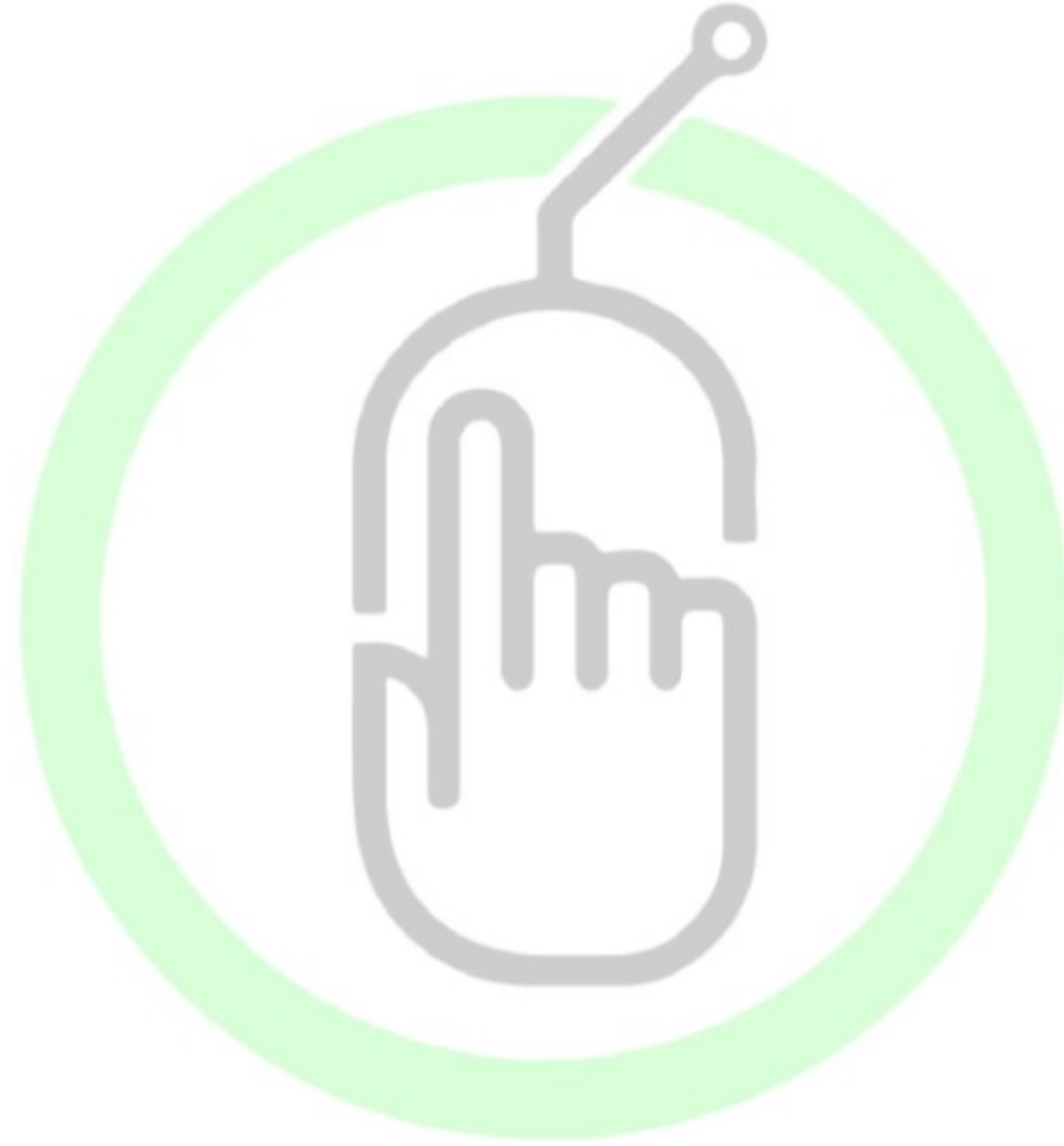
$$\Rightarrow 9 - 16p < 0$$

$$\Rightarrow -16p < -9$$

$$\Rightarrow p < \frac{9}{16} \quad \text{Ans}$$

$$D = 0$$

$$D < 0$$



$$ax^2 + bx + c = 0$$

*** ১৪। $bx^2 + cx + c = 0$ এর মূলদ্বয় α, β হলে দেখাও যে, $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{c}{b}} = 0$

বাকশিৰো- ১৯৮২, ৮৪, ৮৭, ৯০, ৯৫, ০৯'পর্যন্ত

→ $\alpha + \beta = -\frac{c}{b}$
 $\alpha\beta = \frac{c}{b}$

↓ Midterm ***

$$01907-386116$$

L.H.S =

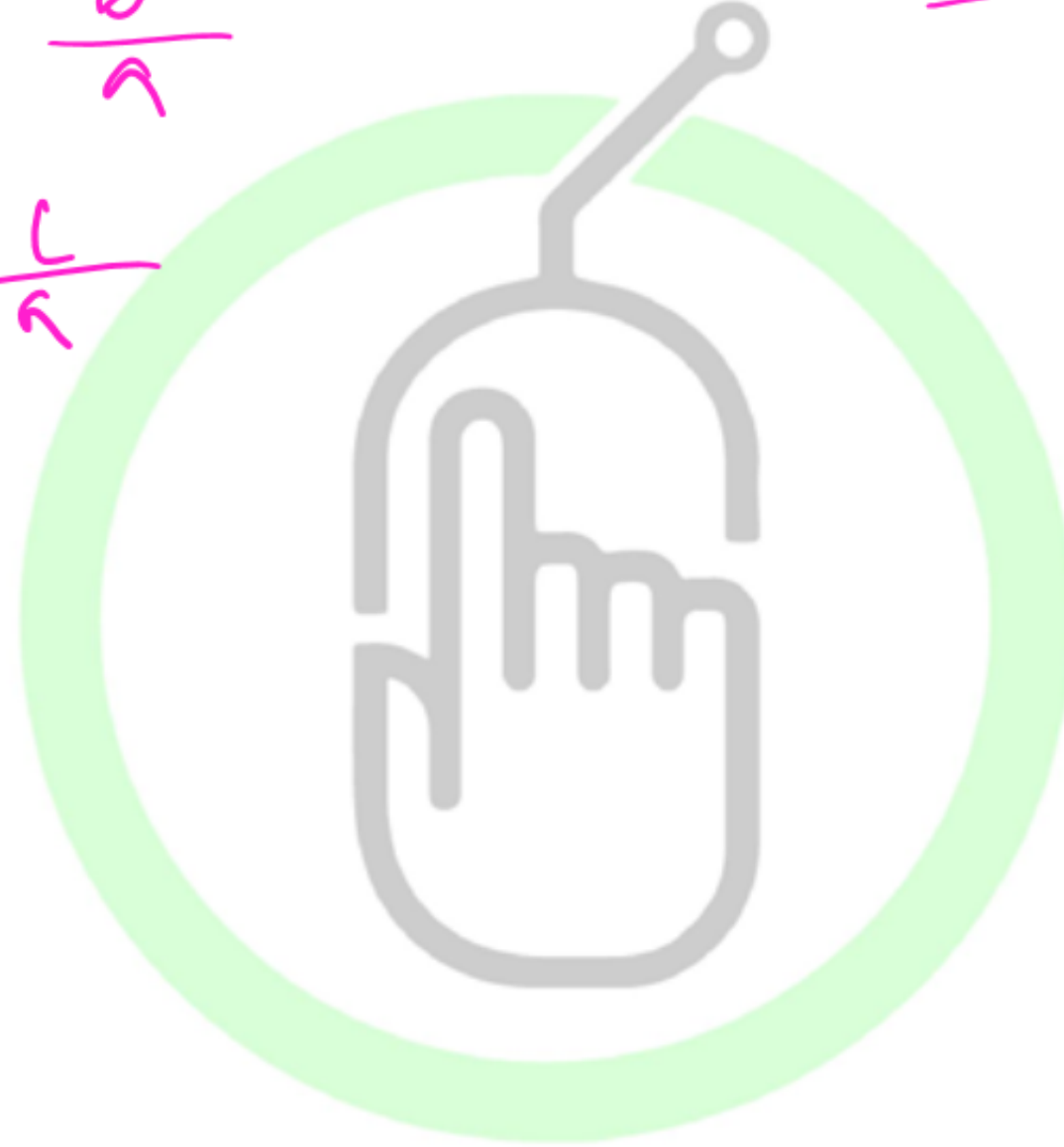
বাকশিৰো problem →

** ২। $ax^2+bx+c=0$ সমীকরণের মূলদ্বয় α ও β হলে, $\alpha + \frac{1}{\alpha}$, $\beta + \frac{1}{\beta}$ মূলগুলো দ্বারা গঠিত সমীকরণটি নির্ণয় কর।

বাকশিবো-০৬, ১৫

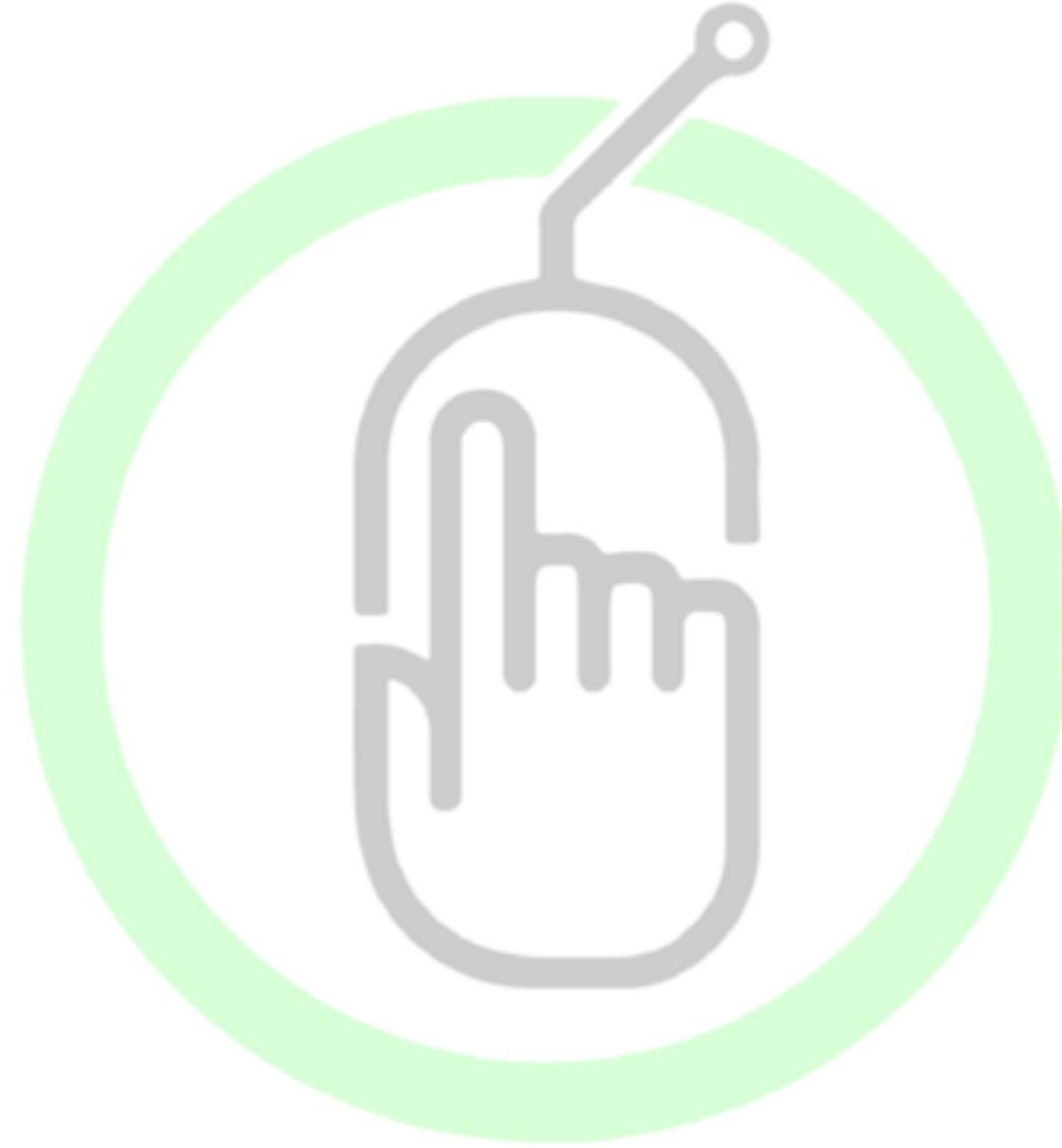
$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = -\frac{c}{a}$$



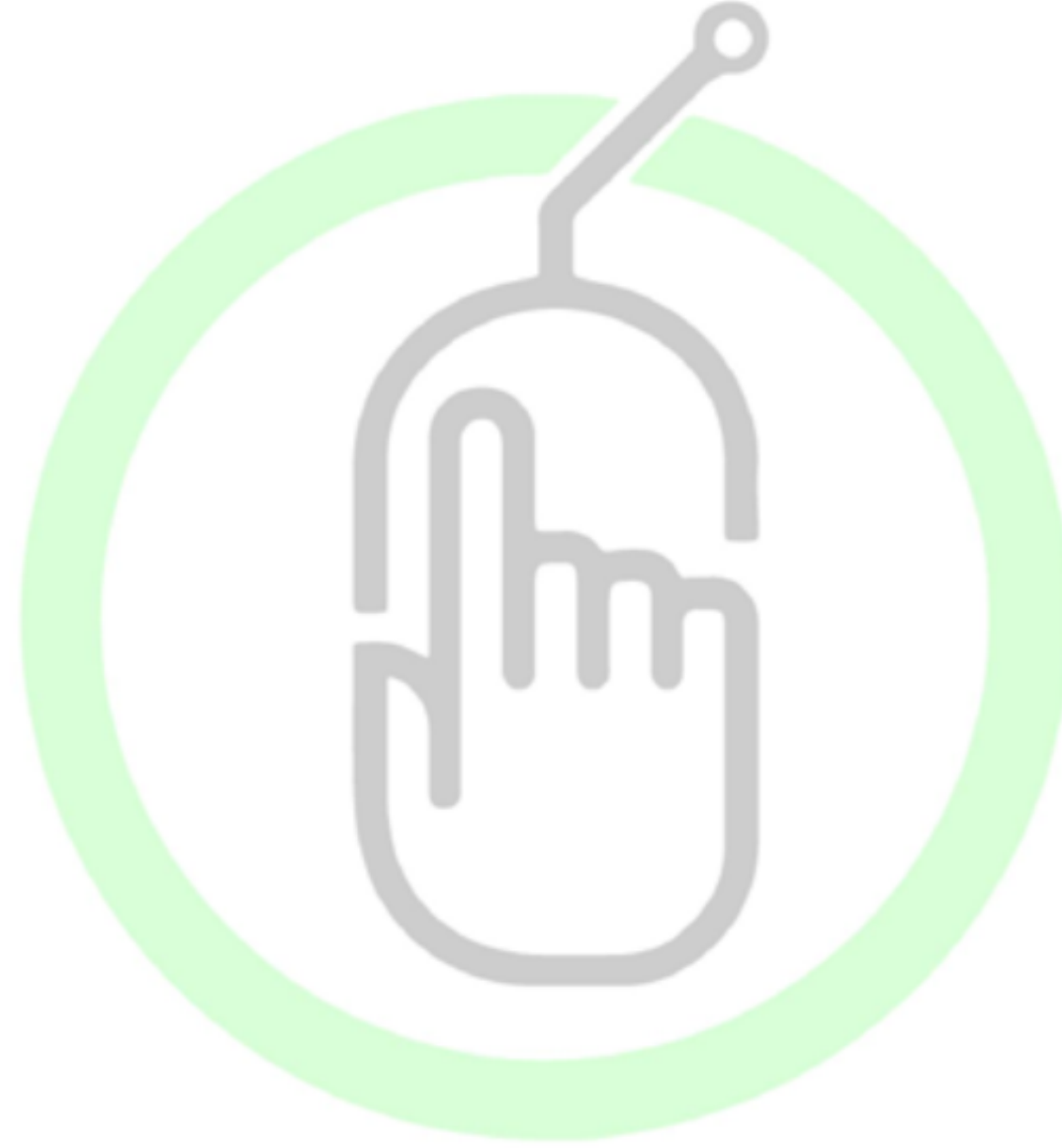
*** ৩। $x^2 + px + q = 0$ সমীকরণের মূলদ্বয় α, β হলে $(\alpha - \beta)^2$ এবং $(\alpha + \beta)^2$ মূলবিশিষ্ট সমীকরণ নির্ণয় কর।

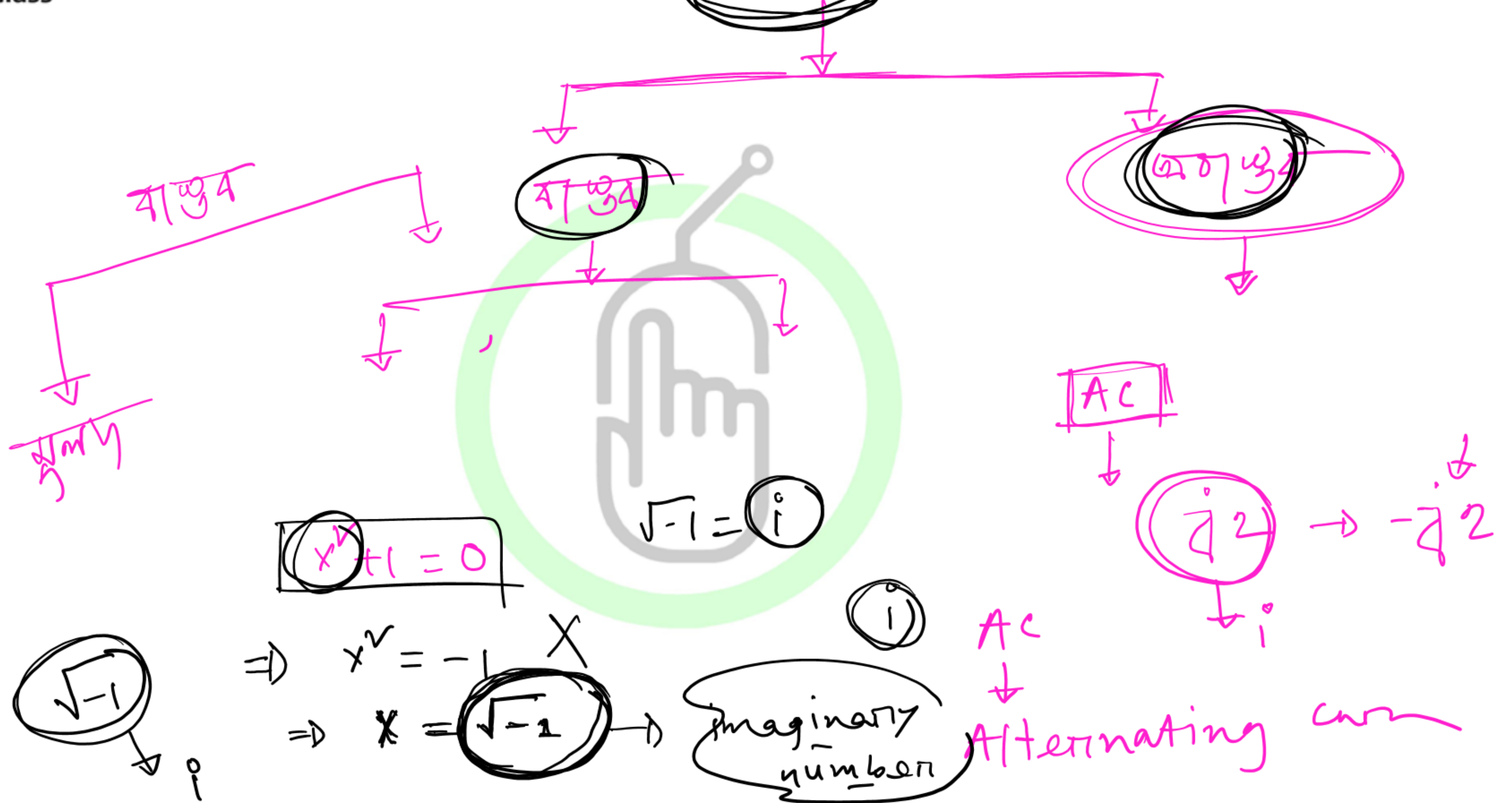
বাকশির্বো- ২০০৬'পরি, ০৭, ০৯, ১৩



*** ৯। যদি $4x^2 + 2x - 1 = 0$ সমীকরণের একটি মূল α হয়, তবে প্রমাণ কর যে, এর অপর মূলটি $4\alpha^3 - 3\alpha$ হবে।

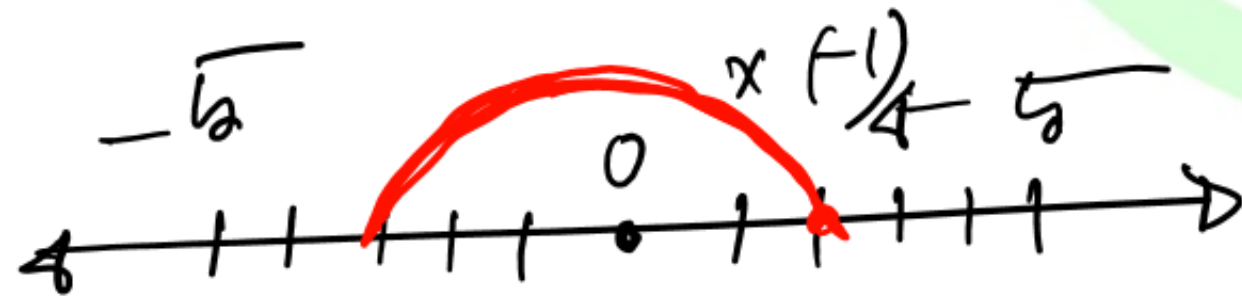
বাকশিবো- ১৯৮৭, ৮৮, ৮৯, ২০০৩'পর, ১৪, ২২





$$\begin{aligned}
 i^0 &= 1 \\
 i^1 &= i \\
 i^2 &= (\sqrt{-1})^2 = -1 \\
 i^3 &= i \cdot i^2 = i \cdot (-1) = -i \\
 i^4 &= i^2 \cdot i^2 = (-1) \cdot (-1) = 1
 \end{aligned}$$

i का चक्र क्या है ?



$\frac{\pi}{2} \times (-1) \rightarrow 180^\circ$ घूर्णन

$$\begin{aligned}
 i^2 &= (\sqrt{-1})^2 \\
 &= -1
 \end{aligned}$$

$$\begin{aligned}
 & \text{Diagram showing } 90^\circ \text{ and } 180^\circ \text{ rotations.} \\
 & \boxed{1} \times \boxed{i} \times \boxed{i} = \boxed{-1} \\
 & \Rightarrow i^2 = -1 \\
 & \Rightarrow i = \sqrt{-1} \\
 & \text{Diagram showing } i^0 = 1
 \end{aligned}$$

$$a \pm bi \rightarrow$$

$$\begin{matrix} \text{real part} \\ a + bi \\ a - bi \end{matrix}$$

imaginary part.

$$\boxed{a + bi} \leftarrow \text{complex number}$$

$$\boxed{7 + 0i}$$

$$7 - 0i$$

$$\begin{matrix} 0 + 7i \\ 0 - 7i \end{matrix}$$

$$\boxed{7i}$$

7

$$\begin{matrix} 7 + 0i \\ \downarrow \quad \downarrow \\ \text{real part} \quad \text{imaginary part} \end{matrix}$$

সবকিছু সমাধান

$$\sqrt[3]{1} = x$$

$$\Rightarrow (\sqrt[3]{1})^3 = x^3$$

$$\Rightarrow 1 = x^3$$

$$\Rightarrow x^3 - 1 = 0$$

$$\Rightarrow (x-1)(x^2+x+1) = 0$$

$$= 2 \text{ cases, } x-1 = 0$$

$$\Rightarrow x = 1$$

$$\sqrt{-9} = \sqrt{3 \cdot (-3)} = \sqrt{3} \cdot \sqrt{-1} = \sqrt{3} \cdot i$$

$$x^3 - 1 = (x-1)(x^2+x+1)$$

$$a^3 - b^3 = (a-b)(a^2+ab+b^2)$$

$$x^2 + x + 1 = 0 \rightarrow ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$= \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

$$= \frac{-1 \pm \sqrt{3}i}{2}$$

$$x = \frac{-1 + \sqrt{3}i}{2}$$

$$x = \frac{-1 - \sqrt{3}i}{2}$$

$(\omega^2)^2 = \omega$
 $\Rightarrow \omega^4 = \omega$
 $\Rightarrow \omega^3 \cdot \omega = \omega$
 $\Rightarrow \omega^3 = 1$

(1,)

$$\frac{-1 + \sqrt{3}i}{2}$$

$$\frac{-1 - \sqrt{3}i}{2}$$

$$(-1)^r = 1$$

$$= \frac{-1 - \sqrt{3}i}{2}$$

$$\frac{-1 + \sqrt{3}i}{2}$$

ω

ω^2

$$1 - 2\sqrt{3}i + 3i^2$$

$$1 - 2\sqrt{3}i + 3(-1)$$

$$-2 - 2\sqrt{3}i$$

$$-2 - 2\sqrt{3}i$$

$$= \frac{-1 - \sqrt{3}i}{2}$$

$$= \frac{(-1 + \sqrt{3}i)^2}{4}$$

$$= \frac{(-1)^2 + 2(-1)(\sqrt{3}i) + (\sqrt{3}i)^2}{4}$$

$$\omega = \frac{-1 + \sqrt{3}i}{2} + \omega^2 = \frac{-1 - \sqrt{3}i}{2}$$

$$= \frac{(-1 + \sqrt{3}i) + (-1 - \sqrt{3}i)}{2}$$

$$= \frac{-2}{2} = -1$$

$$\boxed{\omega + \omega^2 = -1}$$

$$\Rightarrow \boxed{1 + \omega + \omega^2 = 0}$$

$$\boxed{\omega^3 = 1}$$

$$\boxed{1 + \omega + \omega^2 = 0}$$

❖ জটিল সংখ্যার বর্গমূল :

মনে করি, $\sqrt{a + ib} = x + iy$

$$\text{বা, } a + ib = x^2 + 2ixy + i^2y^2 \quad [\text{উভয়পক্ষকে বর্গ করে}]$$

$$\text{বা, } a + ib = x^2 - y^2 + 2ixy \quad [i^2 = -1]$$

বাস্তব ও কাল্পনিক অংশ সমাকৃত করে পাই,

$$x^2 - y^2 = a \dots \dots \dots (i)$$

$$\text{এবং } 2xy = b \dots \dots \dots (ii)$$

$$\text{আবার, } (x^2 + y^2) = (x^2 - y^2) + 4x^2y^2 = (x^2 - y^2) + (2xy)^2 = a^2 + b^2$$

$$\text{বা, } x^2 + y^2 = \sqrt{a^2 + b^2} \dots \dots \dots (iii)$$

(iii) ও (i) নং সমীকরণ যোগ করে পাই,

$$2x^2 = \sqrt{a^2 + b^2} + a$$

$$\text{বা, } x^2 = \frac{1}{2}(\sqrt{a^2 + b^2} + a)$$

$$\therefore x = \pm \frac{1}{\sqrt{2}}(\sqrt{\sqrt{a^2 + b^2} + a})$$

(iii) নং হতে (i) নং সমীকরণ বিয়োগ করে পাই,

$$2y^2 = \sqrt{a^2 + b^2} - a$$

$$\text{বা, } y^2 = \frac{1}{2}(\sqrt{a^2 + b^2} - a)$$

$$\therefore y = \pm \frac{1}{\sqrt{2}}(\sqrt{\sqrt{a^2 + b^2} - a})$$

$$\therefore \text{নির্ণেয় বর্গমূল : } x + iy = \pm \left[\frac{1}{\sqrt{2}}(\sqrt{\sqrt{a^2 + b^2} + a} + i\sqrt{\sqrt{a^2 + b^2} - a}) \right]$$

$$i = \sqrt{-1}$$

*** i) $-8-6\sqrt{-1}$

বাকশিবো-১২, ১৩, ১৪, ১৫, ২২

এটি, $-8-6\sqrt{-1}$

$$\boxed{-8-6i} = \boxed{a+bi}$$

$$(5+2i)^2$$

$$\Rightarrow \boxed{-8-6i} = (a+bi)$$

$$= a^2 + 2a \cdot bi + (bi)^2$$

$$= \boxed{a^2} + \boxed{2abi} - \boxed{b^2}$$

$$= ()$$

$$= 5^2 + 2 \cdot 5 \cdot 2i + (2i)^2$$

$$= 25 + 20i + \boxed{4i^2}$$

যাওক ও সমস্যাটিকে আমরা পরীক্ষা করে = $\boxed{21+20i}$

$$\boxed{a^2 - b^2 = -8}$$

$$2ab = -6$$

$$ab = -3$$

$$(5+2i)^2 = 21+20i$$

$$\boxed{5+2i} = \boxed{21+20i}$$

$$(a+b)^r = (a-b)^r + 4ab$$

$$(a^r + b^r)^r = (a^r - b^r)^r + 4a^r b^r$$

$$= (\quad)$$

अमर की
श्रीम कृष्ण

$$\pm \left(\underset{\uparrow a}{1} - \underset{\uparrow b}{3i} \right)$$

$$a, b$$

$$\begin{matrix} a & \rightarrow & 2 \\ b & \rightarrow & -1 \end{matrix}$$

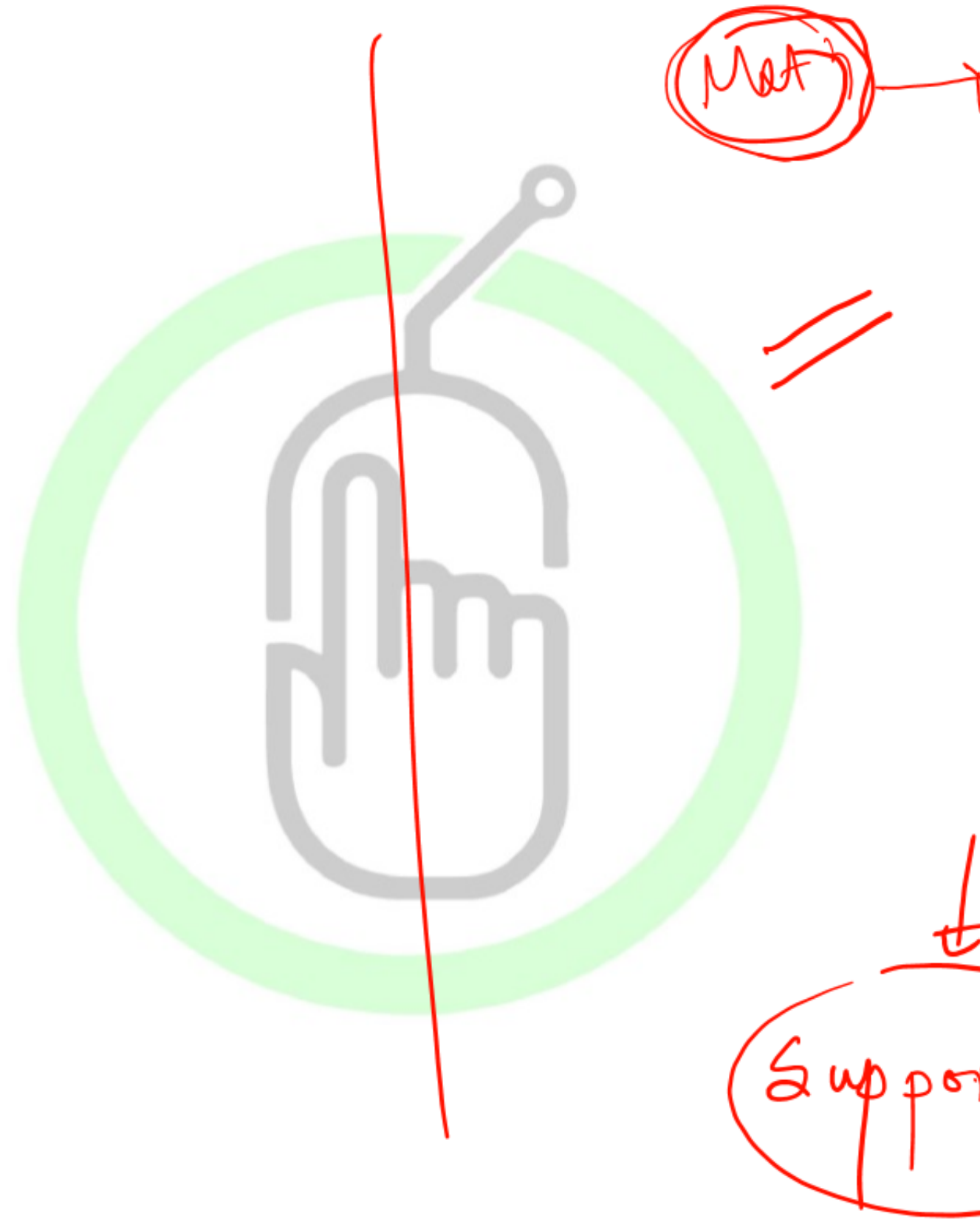
$$\begin{aligned} a^r - b^r &= -8 \\ a^r + b^r &= 2 \end{aligned}$$

$$-2b^r = -10$$

$$b^r = 5$$

$$b = \pm 5$$

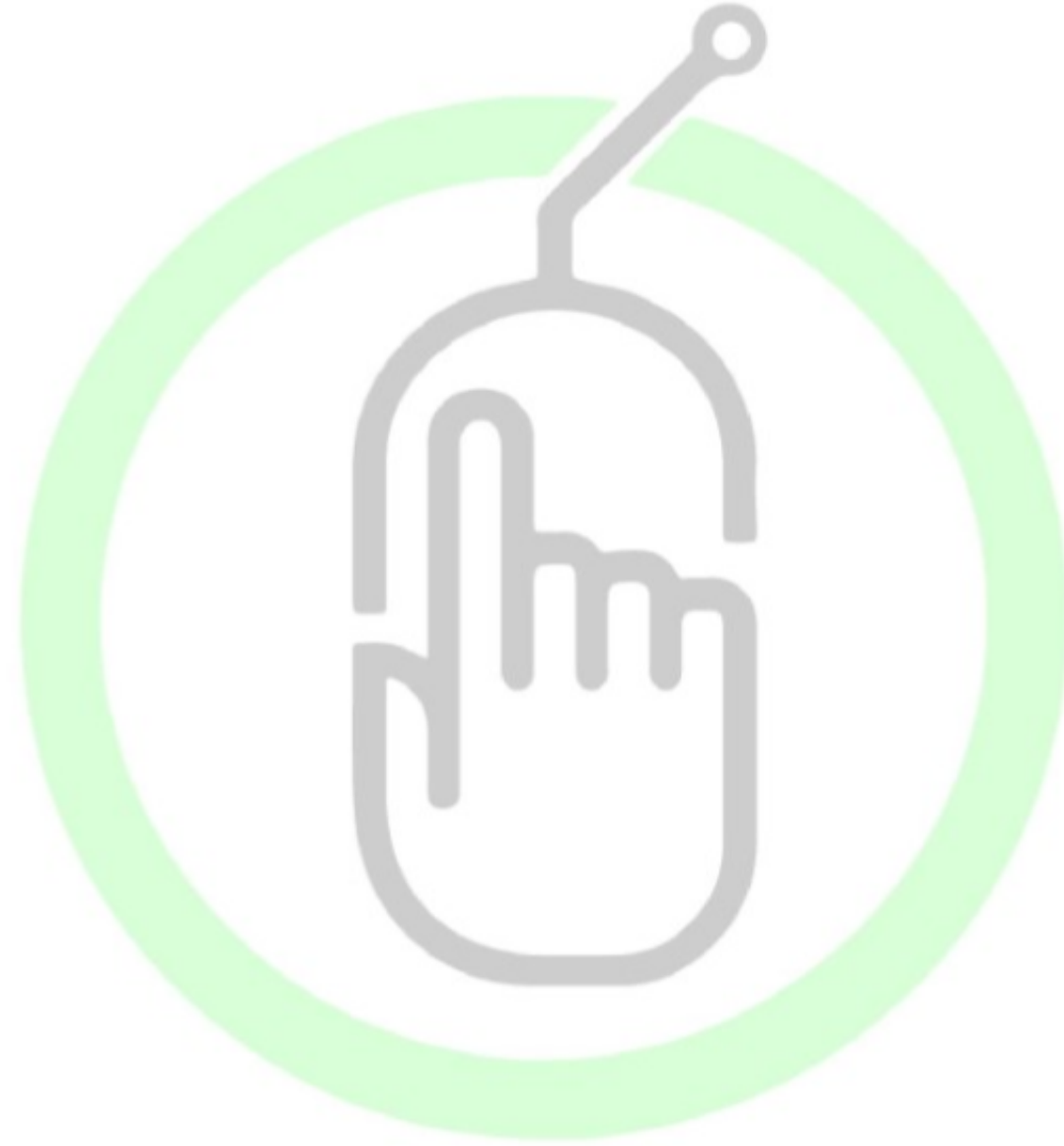
$$\begin{matrix} a + bi \\ \uparrow \quad \uparrow \end{matrix}$$

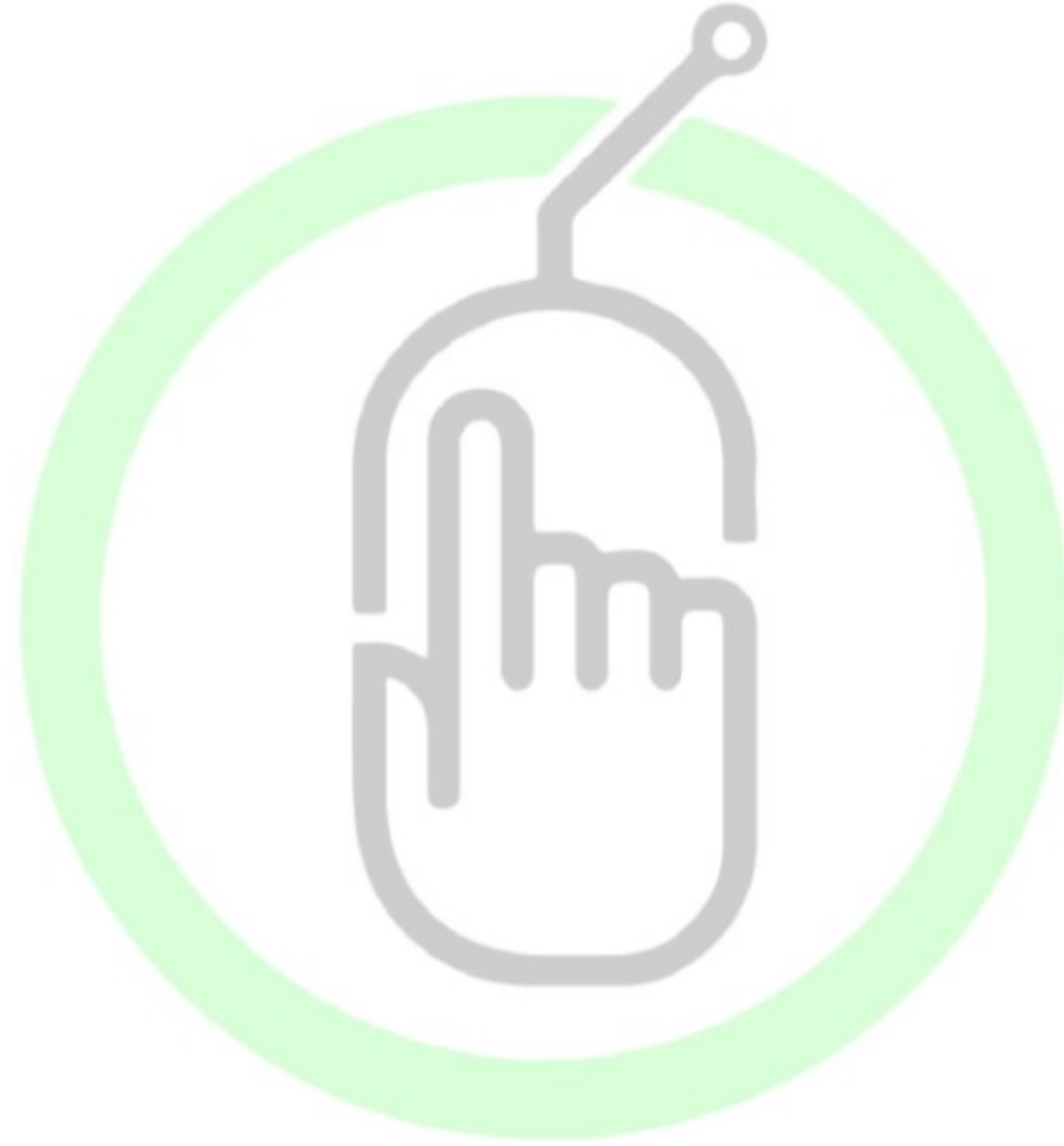


App → class X
Live → X

↓

✓







Practice Class

*** ৩। দেখাও যে, $(x+y)^2 + (x\omega + y\omega^2)^2 + (x\omega^2 + y\omega)^2 = 6xy$
অথবা, দেখাও যে, $(a+b)^2 + (a\omega + b\omega^2)^2 + (a\omega^2 + b\omega)^2 = 6ab$

বাকশিবো- ১১, ১২, ১৫, ২০
বাকশিবো-১১'পরি, ১৫

